








Advanced Time Series Analysis and Forecasting

Chapter 7: Regime-switching models



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **BET**: Bucharest Exchange Trading (index)
- **BFGS**: Broyden–Fletcher–Goldfarb–Shanno (quasi-Newton optimiser)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **CHP**: Carrasco–Hu–Ploberger (test, 2014)
- **CRPS**: Continuous Ranked Probability Score
- **DLW**: Diebold–Lee–Weinbach (TVTP model, 1994)
- **ECB**: European Central Bank
- **ECM**: Error Correction Model
- **EM**: Expectation–Maximisation (algorithm)
- **EODHD**: EOD Historical Data (market-data provider)
- **ES**: Expected Shortfall
- **ESI**: Economic Sentiment Indicator (European Commission)
- **FFBS**: Forward Filtering, Backward Sampling
- **FRED**: Federal Reserve Economic Data
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **GNP**: Gross National Product
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HICP**: Harmonised Index of Consumer Prices
- **HMP**: Haas–Mittnik–Paolella (MS-GARCH, 2004)
- **HQ**: Hannan–Quinn (information criterion)

Acronyms used in this material (2/2)

- **KS**: Kolmogorov–Smirnov (test)
- **LLM**: Large Language Model
- **LR**: Likelihood Ratio (test)
- **MCMC**: Markov Chain Monte Carlo
- **MFM**: Modelling Financial Markets
- **ML**: Machine Learning
- **MS**: Markov Switching
- **MS-GARCH**: Markov-Switching GARCH
- **MS-VAR**: Markov-Switching Vector AutoRegression
- **MSI**: Markov-Switching Intercept
- **MSI-AR**: Markov-Switching Intercept AutoRegression
- **MSIAH**: Markov-Switching Intercept, Autoregressive parameters and Heteroskedasticity
- **MSIH**: Markov-Switching Intercept and Heteroskedasticity
- **MSIH-AR**: Markov-Switching Intercept and Heteroskedasticity AutoRegression
- **MSM**: Markov-Switching Mean (Krolzig notation)
- **MSM-AR**: Markov-Switching Mean AutoRegression
- **NBER**: National Bureau of Economic Research
- **PIT**: Probability Integral Transform
- **QPS**: Quadratic Probability Score
- **RMSE**: Root Mean Squared Error
- **ROL**: Romanian old leu (ISO 4217 code, until 2005)
- **STAR**: Smooth Transition AutoRegression
- **SWARCH**: Switching ARCH (Hamilton–Susmel)
- **TAR**: Threshold AutoRegression
- **TSA**: Time Series Analysis
- **TVTP**: Time-Varying Transition Probabilities
- **US**: United States
- **VAR**: Vector AutoRegression
- **VAT**: Value Added Tax

Today's question and route

- **Question**: when the data-generating process itself changes (recession and expansion, calm and panic, a peg and a float), how do we infer the regime, estimate the model, test it and forecast with it?
  - the regime is never observed: everything rests on its probability given the data
- **Route** of the chapter
  - the Hamilton filter and the Kim smoother derived; EM and numerical maximum likelihood; identification and label switching
  - testing the number of regimes; time-varying transition probabilities; Markov-switching VAR; Markov-switching GARCH
  - Bayesian estimation by Gibbs sampling; regimes, long memory and breaks; forecasting and its evaluation
  - applications: US business cycles, Romanian inflation and GDP, S&P 500 and BET, EUR/RON
- We build on TSA, Chapter 10 (Hamilton's model with statsmodels), Chapter 2 (threshold models, breaks) and Chapter 6 (state space, Bayesian filtering); Seminar 7 comes before this lecture

Learning outcomes

- Derive the Hamilton filter, the likelihood and the Kim smoother, and write them in numpy with a numerically stable recursion
- Estimate Markov-switching regressions, autoregressions and VARs by EM and by numerical maximum likelihood, and diagnose local and degenerate maxima
- Explain why the likelihood-ratio test for the number of regimes is non-standard, and test by bootstrap and by information criteria
- Specify time-varying transition probabilities, regime-dependent impulse responses and MS-GARCH models without path dependence
- Estimate a switching model by Gibbs sampling, handle label switching, and evaluate regime forecasts with scores and probability scores

Reading, data and tools

- Backbone: Hamilton (1994), Ch. 22; Kim and Nelson (1999); Krolzig (1997); Frühwirth-Schnatter (2006)
  - Surveys: Hamilton (2016); Ang and Timmermann (2012)
- Python Quantlets of this chapter: Quantlets/Ch_07
  - Hamilton filter, Kim smoother, EM, expanded-state ML, TVTP, MS-VAR, MS-GARCH and the Gibbs sampler written in `numpy`
  - comparison: `statsmodels` `MarkovRegression` and `MarkovAutoregression`
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Use
US real GNP growth, 1951Q2–1984Q4 (Hamilton's data); US industrial production and leading indicator, 1948–1991 (Filardo's data)	data sets distributed with statsmodels	replication
US real GDP, unemployment rate, NBER recession quarters	FRED (GDPC1, UNRATE, USRECQ)	dating, MS-VAR, forecasts
Romanian real GDP (quarterly) and HICP (monthly)	Eurostat (namq_10_gdp, prc_hicp_minr)	Gibbs, inflation regimes
S&P 500, BET: daily closes	EODHD (data/market)	MS-GARCH, bull/bear volatility regimes
EUR/RON	ECB reference rate to June 2005, BNR reference rate from July 2005	

- All sources are public and need no account or key

Origins of regime-switching models



Andrey Markov (1856–1922), around 1886

Photo: unknown author (1886); public domain; Wikimedia Commons

- 1906–1913: Markov chains, a dependence that needs only the present state
- 1970: Baum et al. (1970): maximum likelihood for hidden Markov models (the Baum–Welch algorithm, an EM before EM); speech recognition Rabiner (1989)
- 1973: Goldfeld and Quandt (1973): switching regressions in econometrics
- 1989–1990: Hamilton (1989) dates US recessions from GNP alone; Hamilton (1990) gives the EM algorithm; Kim (1994) the smoother
- 1993–2006: Bayesian estimation Albert and Chib (1993), Chib (1996), Frühwirth-Schnatter (2001); MS-VAR Krolzig (1997); MS-GARCH Gray (1996), Haas, Mittnik and Paolella (2004)

Prior knowledge from TSA and new material

- Known (TSA, Chapter 10): $y_t = \mu_{S_t} + \varepsilon_t$, transition matrix, expected durations, ergodic probabilities, filtered and smoothed probabilities from `statsmodels`
  - US recessions from GDP, Romanian growth regimes, the effect of 2020
- New: every step derived and coded; estimation as an optimisation problem with traps; inference on the number of regimes
  - extensions: transitions that depend on covariates, systems (MS-VAR), volatility dynamics (MS-GARCH), Bayesian estimation
- Chapter 2 split the sample by an observed threshold (TAR, STAR) or a date (breaks); here the switch is a latent Markov chain

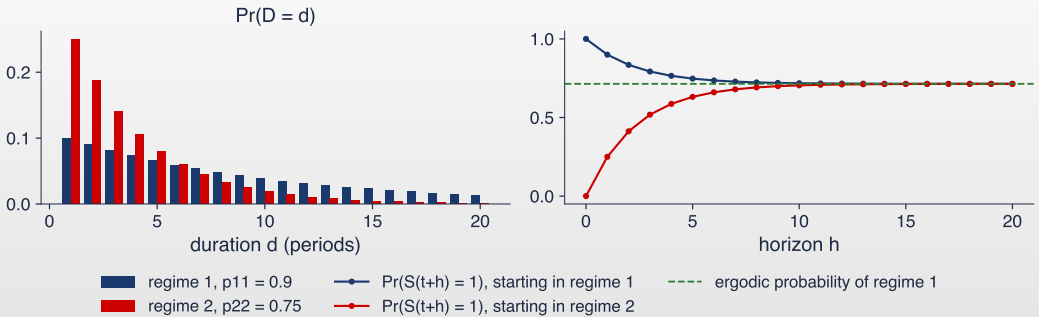
The model family (Krolzig's notation)

- MS(K)-AR(p): $y_t = \nu_{S_t} + \sum_{k=1}^{p} \phi_{k,S_t} y_{t-k} + \sigma_{S_t}\varepsilon_t$, $\varepsilon_t \sim$ i.i.d. $N(0,1)$, $S_t \in \{1,\dots,K\}$
  - S_t: the regime at t, one of K values; ν_{S_t}, ϕ_{k,S_t}, σ_{S_t}: the intercept, the AR(p) coefficients and the shock s.d. of the current regime
  - S_t a homogeneous Markov chain, $p_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$, independent of ε Krolzig (1997); $\mathrm{P} = (p_{ij})$: the transition matrix, rows sum to 1
- Which parameters switch gives the name:
  - **MSI**: intercept ν; **MSIH**: intercept and variance; **MSIAH**: intercept, AR coefficients and variance
  - **MSM** (Hamilton 1989): the *mean* μ_{S_t} switches, $y_t - \mu_{S_t} = \sum_k \phi_k (y_{t-k} - \mu_{S_{t-k}}) + \sigma\varepsilon_t$
- MSI: after a switch the mean adjusts gradually; MSM: it jumps at once; MSM needs the regimes of $p+1$ periods
- Special cases: $p_{ij} = \pi_j$ for all i is an i.i.d. mixture; an upper-triangular P is a change-point model Chib (1998)

The chain: ergodicity, durations, persistence

- Ergodic probabilities: $\boldsymbol{\pi}'\mathrm{P} = \boldsymbol{\pi}'$, $1'\boldsymbol{\pi} = 1$; with $K = 2$, $\pi_1 = (1 - p_{22})/(2 - p_{11} - p_{22})$
  - $\boldsymbol{\pi}$: the long-run share of time spent in each regime; 1: a vector of ones
  - durations are geometric: $\Pr(D_i = d) = p_{ii}^{d-1}(1 - p_{ii})$, $\mathbb{E}D_i = 1/(1 - p_{ii})$ (memoryless); D_i: the number of periods a visit to regime i lasts
- AR(1) representation Hamilton (1994), §22.2: $\xi_t = 1\{S_t = 1\}$ satisfies $\xi_t = (1 - p_{22}) + \lambda\xi_{t-1} + v_t$, $\lambda = p_{11} + p_{22} - 1$
  - $1\{\cdot\}$: the indicator (1 if true, 0 otherwise); v_t: the innovation; it is a martingale difference, not independent of the past (its variance depends on S_{t-1})
  - $\Pr(S_{t+h} = 1 \mid S_t) \to \pi_1$ at the rate λ^h: λ is the persistence of the regime process
- Duration dependence (recessions that are more likely to end the longer they last) needs an extended state or a semi-Markov chain Maheu and McCurdy (2000)

Durations and the speed of mixing



- $p_{11} = 0.9$, $p_{22} = 0.75$: expected durations 10 and 4 periods; $\pi_1 = 0.714$; $\lambda = 0.65$
- The gap to the ergodic probability shrinks by the factor λ each period: after 10 periods the starting regime is almost forgotten

ATS_ch7_hamilton

Derivation of the Hamilton filter

- Notation: $Y_t = (y_1, \ldots, y_t)$, $\hat{\xi}_{t|s}$ the K-vector $\Pr(S_t = j \mid Y_s)$, η_t the vector of densities $f(y_t \mid S_t = j, Y_{t-1})$
  - **prediction** (law of total probability, Markov property): $\hat{\xi}_{t|t-1} = \mathrm{P}' \hat{\xi}_{t-1|t-1}$
  - **update** (Bayes): $\hat{\xi}_{t|t} = \dfrac{\hat{\xi}_{t|t-1} \odot \eta_t}{1'(\hat{\xi}_{t|t-1} \odot \eta_t)}$
  - **likelihood**: $f(y_t \mid Y_{t-1}) = 1'(\hat{\xi}_{t|t-1} \odot \eta_t)$, so $\ln L = \sum_t \ln f(y_t \mid Y_{t-1})$
- Start: $\hat{\xi}_{1|0} = \boldsymbol{\pi}$ (ergodic) or a free parameter; $\odot$ is the element-wise product
- Same structure as the Kalman filter (Chapter 6): predict, update, prediction-error decomposition; here the state is discrete and the update is exact

Numerics and cost

- Densities underflow (e.g. 10^{-300} after a large outlier): work with $\ln \eta_t$ and subtract $m_t = \max_j \ln \eta_{jt}$
  - $\ln f(y_t \mid Y_{t-1}) = m_t + \ln \sum_j \hat{\xi}_{j,t|t-1} e^{\ln \eta_{jt} - m_t}$ (log-sum-exp); the filtered vector is unchanged
- Cost: $O(TK^2)$; with MSM-AR(p) the state is $(S_t, \ldots, S_{t-p})$, $M = K^{p+1}$ (32 for Hamilton's $K = 2$, $p = 4$)
  - the expanded transition matrix is sparse: from $(i_0, \ldots, i_p)$ only to $(j, i_0, \ldots, i_{p-1})$, with probability $p_{i_0 j}$
  - MSI-AR(p) needs only K states: the lagged y enter as regressors
- Our numpy filter and `statsmodels` give the same log-likelihood to machine precision when the initial distribution is the same

Derivation of the Kim smoother

- Goal: $\hat{\xi}_{t|T}$, the regime probabilities given the whole sample Kim (1994)
  - key step: $\Pr(S_t = i \mid S_{t+1} = j, Y_T) = \Pr(S_t = i \mid S_{t+1} = j, Y_t)$: given S_{t+1}, future y carry no extra information on S_t
  - exact for MSI/MSIH/MSM on the expanded state; an approximation only when the state also has a continuous part (Kim 1994, Chapter 6)
  - proof of the key step: Appendix
- Bayes on the pair: $\Pr(S_t = i, S_{t+1} = j \mid Y_T) = \dfrac{\hat{\xi}_{i,t|t}\, p_{ij}}{\hat{\xi}_{j,t+1|t}}\, \hat{\xi}_{j,t+1|T}$
  - sum over j: $\hat{\xi}_{t|T} = \hat{\xi}_{t|t} \odot \left[\mathrm{P}(\hat{\xi}_{t+1|T} \oslash \hat{\xi}_{t+1|t})\right]$, backwards from $\hat{\xi}_{T|T}$; $\oslash$: element-wise division
- The joint probabilities of (S_t, S_{t+1}) are the E-step of EM; sampling instead of summing gives FFBS (section Bayesian estimation)

Predicted, filtered and smoothed regime probabilities

- **Predicted** $\hat{\xi}_{t+1|t}$: what a forecaster expects for next period; it enters the forecast density
- **Filtered** $\hat{\xi}_{t|t}$: what could be known at t; the real-time signal (Chauvet and Piger 2008)
  - evaluate real-time dating with filtered probabilities and data vintages, never with smoothed ones
- **Smoothed** $\hat{\xi}_{t|T}$: the historian's answer; it uses the future and is revised as data arrive
  - the dating of past regimes, and the weights of the M-step
- Confusing them is the most common error in applied work: a smoothed probability is not a forecast

EM: the complete-data likelihood

- If $S_1, \ldots, S_T$ were observed, the log-likelihood would split Dempster, Laird and Rubin (1977), Hamilton (1990):
  - $\ln L_c = \sum_j 1\{S_1 = j\} \ln \rho_j + \sum_{t \geq 2} \sum_{i,j} 1\{S_{t-1} = i, S_t = j\} \ln p_{ij} + \sum_t \sum_j 1\{S_t = j\} \ln \eta_{jt}(\theta_j)$
  - $\rho_j = \Pr(S_1 = j)$: initial probabilities; $\eta_{jt}(\theta_j)$: the density of y_t in regime j, with parameters θ_j
  - transition counts and regime-by-regime regressions: closed forms
- **E-step**: replace the indicators by their expectations given Y_T and the current $\theta^{(m)}$ (m: the iteration)
  - $\hat{\xi}_{j,t|T}$ and $\hat{\xi}_{ij,t|T} = \Pr(S_{t-1} = i, S_t = j \mid Y_T)$: exactly the output of the Kim smoother
- Baum et al. (1970) derived the same recursion for hidden Markov models (the forward–backward algorithm)

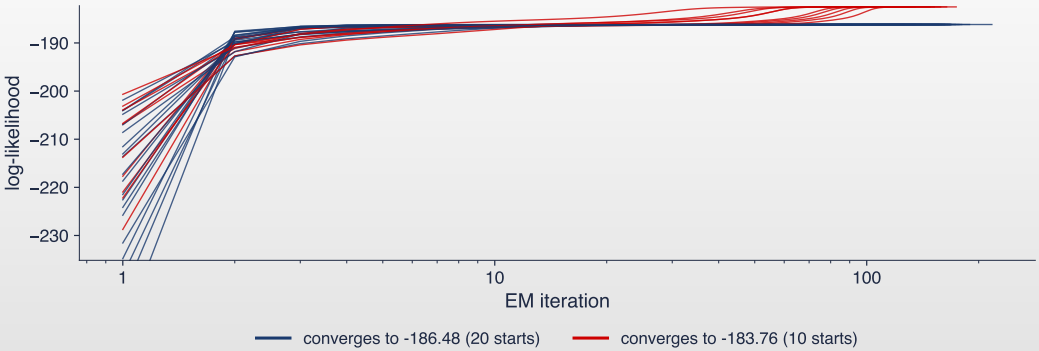
EM: the M-step in closed form

- Transitions: $p_{ij}^{(m+1)} = \dfrac{\sum_{t=2}^{T} \hat{\xi}_{ij,t|T}}{\sum_{t=2}^{T} \hat{\xi}_{i,t-1|T}}$ (expected transitions over expected visits); $\rho^{(m+1)} = \hat{\xi}_{1|T}$
- Switching coefficients (MSIAH): weighted least squares $\beta_j^{(m+1)} = (X' W_j X)^{-1} X' W_j y$, $W_j = \mathrm{diag}(\hat{\xi}_{j,t|T})$; X: rows x_t' (constant and lags), y: the observations; each date weighs by its probability of being in regime j
- Variances: $\sigma_j^{2(m+1)} = \sum_t \hat{\xi}_{j,t|T} (y_t - x_t'\beta_j)^2 / \sum_t \hat{\xi}_{j,t|T}$; MS-VAR: the same with matrices
- Common coefficients (MSIH-AR): one stacked weighted regression with weights $\hat{\xi}_{j,t|T}/\sigma_j^2$ (a conditional M-step, ECM)
- MSM-AR: μ and ϕ enter as products, so no closed form: numerical maximisation (or EM with a numerical M-step)
- Derivation of the M-step for P and of the monotonicity of EM: ▸ Appendix

EM and numerical ML in practice

- EM never lowers the likelihood, is robust far from the optimum, but slow near it (linear convergence)
  - monotonicity needs ρ as a free parameter; with $\rho = \boldsymbol{\pi}(\mathrm{P})$ the M-step for P is only approximate
- Numerical ML: BFGS on unconstrained parameters ($\ln \sigma_j^2$, logits of p_{ij}); standard errors from the Hessian by the delta method
  - practice: EM from many starts, then a Newton-type polish and the Hessian
- The likelihood is unbounded: $\sigma_j \to 0$ around one observation with $p_{jj} \to 0$ gives $\ln L \to \infty$
  - remedies: a lower bound on σ_j / σ_k Hathaway (1985), priors (section Bayesian estimation), or reject regimes that last one period
- Standard errors near the boundary ($p_{ii} \approx 1$) are unreliable: report profile likelihoods or bootstrap intervals

EM from 30 starting values



- MSIH(2)-AR(1) on Hamilton's GNP data: switching intercept and variance, common AR(1) coefficient; random starting values; log scale for iterations

ATS_ch7_estimation

Interpreting the EM paths

- 2 limits (log-likelihood, number of starts): -186.48 (20); -183.76 (10); median 159 iterations, at most 216
- The highest (-183.76) is **degenerate**: a regime with $\sigma^2 = 0.049$, $p_{11} < 10^{-9}$ and mean 1.11, used by 29 isolated quarters
- `statsmodels` (20 random searches) reports -185.73, a third local maximum that none of our starts reached
- Lesson: report the starting-value design, all local maxima, and why the chosen one is economically meaningful

Identification and label switching

- The likelihood is invariant to the $K!$ permutations of the labels: $(\theta_1, \theta_2, \mathrm{P})$ and $(\theta_2, \theta_1, \mathrm{P}'')$ fit equally well
  - for ML this is harmless: pick one mode and name the regimes by an ordering ($\mu_1 < \mu_2$ or $\sigma_1 < \sigma_2$)
  - for MCMC it is not: the sampler jumps between modes and posterior means of μ_1 average the regimes Celeux, Hurn and Robert (2000), Stephens (2000)
- Identification also fails locally: K regimes with $\theta_i = \theta_j$, or a regime never visited, leave P unidentified
  - this is the root of the testing problem of the next section
- Statistical regimes need not be economic regimes: the same GDP data give recessions (MSM), volatility eras (MSIH) or single outliers

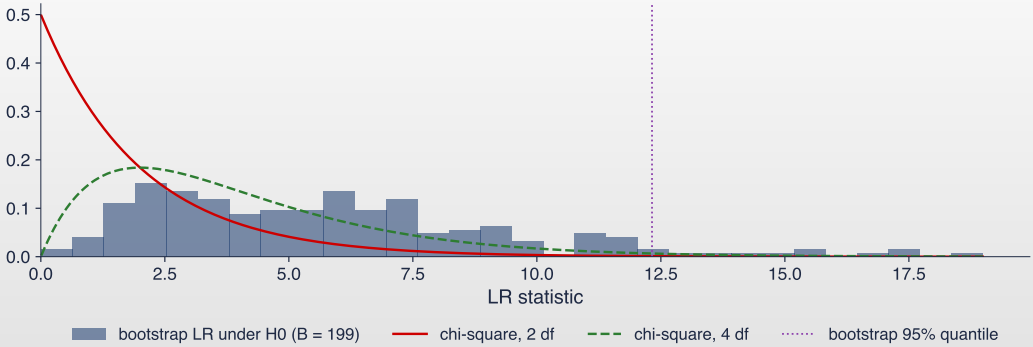
The non-standard LR test

- H_0: one regime ($\mu_1 = \mu_2$, $\sigma_1 = \sigma_2$) against H_1: two regimes
  - (i) under H_0, p_{11} and p_{22} are **not identified**: nuisance parameters present only under H_1, the Davies problem of Chapter 2 Davies (1987)
  - (ii) the null can be written as $p_{11} = 1$: a parameter on the **boundary**
  - (iii) the score with respect to $\mu_2 - \mu_1$ is **identically zero** at H_0: the information matrix is singular
- Each of the three breaks one of the conditions for $2 \ln \Lambda \to \chi^2_q$ (Λ: the likelihood ratio; q: the number of restrictions): counting parameters does not give the critical value
- Same structure as testing linearity against TAR or STAR (Chapter 2), with a latent instead of an observed switch

Solutions in the literature

- Hansen (1992): treat the likelihood as an empirical process in the nuisance parameters and bound the standardised LR; conservative, heavy to compute
  - applied to Hamilton's GNP model: no evidence against one regime at conventional levels
- Garcia (1998): the asymptotic null distribution of the sup-LR for MS models with switching mean (and variance), with tabulated critical values
- Cho and White (2007): quasi-LR against a two-component mixture (an i.i.d. switch has the same null behaviour)
- Carrasco, Hu and Ploberger (2014): an optimal score-type test against Markov switching that needs only the null model
- Practice: **parametric bootstrap** of the LR
  - simulate from the estimated null; re-estimate both models (with several starts) on each sample
  - $p = (1 + \#\{LR^* \geq LR\})/(B + 1)$; LR^*: the B bootstrap statistics; $\#$: their count above LR
- Selection rather than testing: AIC, BIC, HQ, or Bayes factors through the marginal likelihood Chib (1995)

The bootstrap null distribution of the LR statistic



- US GDP growth, 1947Q2–2019Q4 ($T = 290$): H_0 Gaussian AR(1), H_1 MSIH(2)-AR(1); 199 samples simulated from the estimated AR(1), both models re-estimated, EM with 4 starts

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Interpreting the test

- Bootstrap 95% quantile 12.32, against 5.99 (χ^2_2) and 9.49 (χ^2_4); mean of the null draws 5.79: counting parameters gets the critical value wrong
- Observed LR = 71.7, bootstrap p-value 0.005 (the smallest possible with $B = 199$): two regimes
- But which regimes? Standard deviations 0.46 and 1.11 pp, $p_{ii} = 0.990$ and 0.985: volatility eras (the Great Moderation), not recessions
- BIC: 755.5 ($K = 1$), 706.4 ($K = 2$), 734.7 ($K = 3$); AIC: 744.5, 680.8, 687.0

Hamilton (1989): data and model



Chicago, February 1931: the regime every forecaster fears

Photo: US National Archives (1931); public domain; Wikimedia Commons

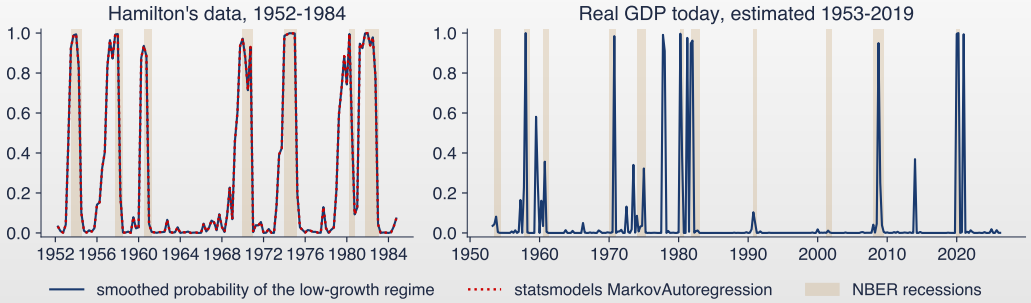
- Hamilton (1989), Section 4: US real GNP, 100 × log change, 1951Q2–1984Q4, MSM(2)-AR(4)
- Equation (4.3): $y_t - \mu_{S_t} = \sum_{k=1}^{4} \phi_k (y_{t-k} - \mu_{S_{t-k}}) + \sigma \varepsilon_t$; estimates in Table I
- The claim: the dates where the low-growth regime is likely match the NBER recessions, although the NBER dates were never used
- We replicate on his data set, compare numpy with `statsmodels`, then re-estimate on today's GDP

Replication: Hamilton's data, two implementations

Estimate	μ_0	μ_1	ϕ_1	ϕ_2	ϕ_3	ϕ_4	σ	$\ln L$
numpy	-0.359	1.164	0.013	-0.058	-0.247	-0.213	0.769	-181.26
(standard error)	(0.265)	(0.075)	(0.120)	(0.138)	(0.107)	(0.111)		
statsmodels	1.164	-0.359	0.013	-0.058	-0.247	-0.213	0.769	-181.26

- Transitions: $p_{00} = 0.755$ (stay in the low regime), $p_{11} = 0.904$; expected durations 4.1 and 10.4 quarters; $T = 131$ after 4 lags
- Largest difference between the two smoothed probability paths: $< 10^{-4}$
- Estimation: BFGS on the expanded state of 32 regime combinations, 8 random starts; standard errors from the numerical Hessian

Hamilton's regimes on the original and on current data



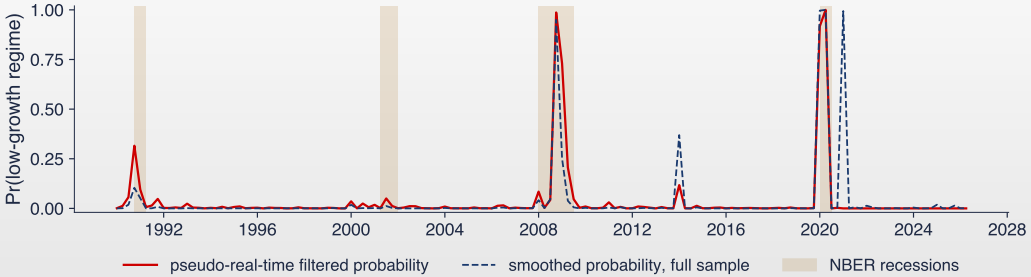
⊡ Left: smoothed probability of the low-growth regime on Hamilton's data. Right: the same specification on today's GDPC1, estimated on 1953Q2–2019Q4 ($T = 267$), probabilities to 2026Q2 with the same parameters

ATS_ch7_hamilton

Interpreting the replication

- On his data, the regime tracks the NBER: QPS 0.179 and 89% concordance (probability above 0.5 against the NBER quarters) Diebold and Rudebusch (1989), Harding and Pagan (2002)
- On today's data the low regime changes nature: mean -1.22%, $p_{00} = 0.26$ (duration 1.4 quarters): single sharp falls, not recessions
- It catches 2008Q4 (0.95) and 2020Q2 (1.00) but not 2001 (at most 0.01); QPS 0.229 on 37 NBER quarters
- After 1984 recessions are rarer and milder (the Great Moderation): a fixed two-mean model loses its grip; this motivates TVTP, MSIH and three regimes

Dating in pseudo real time



- Hamilton's model re-estimated every year on the data available then (current vintage, no data revisions); filtered probability of each quarter with information up to that quarter, 1990–2026

ATS_ch7_hamilton

Interpreting real-time dating

- First quarter above 0.5: 1990–91 none (maximum 0.32), 2001 none (maximum 0.05), 2008 2008Q4, 2020 2020Q1
- QPS before 2020: 0.127 in real time against 0.152 smoothed; 0 false signals above 0.5
- Chauvet and Piger (2008): with real-time vintages, MS models call the start of recessions faster than the NBER announcement, but mild recessions are missed
- Our exercise flatters the model: today's revised GDP was not available in 2001

Letting the chain depend on observables

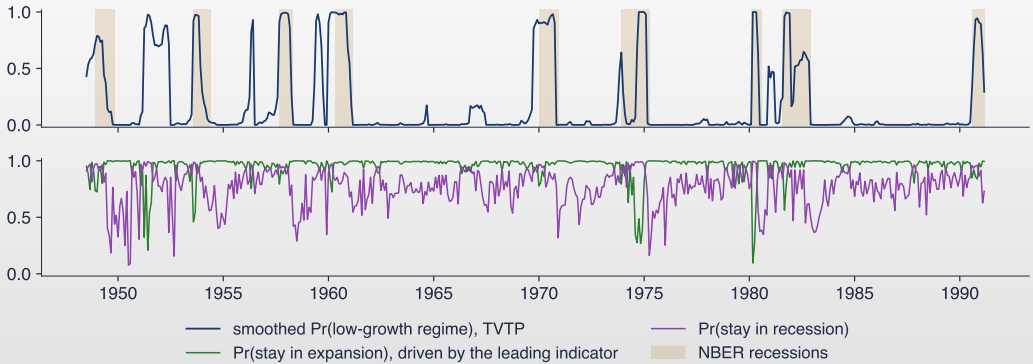
- Diebold, Lee and Weinbach (1994), Filardo (1994): $p_{ii,t} = \Pr(S_t = i \mid S_{t-1} = i, z_{t-1}) = \dfrac{\exp(z'_{t-1}\gamma_i)}{1 + \exp(z'_{t-1}\gamma_i)}$
  - z_{t-1}: observed variables known at $t-1$ (with a constant); γ_i: their coefficients; the logistic function keeps $p_{ii,t}$ in $(0,1)$
  - a leading indicator, the term spread or a policy rate can make a recession more or less likely to begin or end
  - expected durations now vary over time; the chain is no longer homogeneous
- Filter and smoother unchanged, with P_t in place of P; EM: the M-step for γ_i is a weighted logit (DLW)
  - z_{t-1} must be predetermined and must not depend on S_t; if z reacts to the regime, the model is misspecified (endogenous switching)
- Test of constant transitions: H_0: slopes of $\gamma_i = 0$ is a standard LR test, since the regimes exist under both hypotheses

Replication: Filardo (1994)

Model	μ_0	μ_1	γ_0	γ_1	ln L	statsmodels
TVTP, numpy	-0.87	0.50	(1.49; -1.09)	(4.33; 1.75)	-586.20	-586.20
(standard error)			(0.48; 0.62)	(0.76; 0.52)		
constant transitions		2.77	$p_{00} = 0.983$	$p_{11} = 0.14$	-590.67	

- US industrial production growth, monthly, $T = 513$ (Filardo's data); MSM(2)-AR(4); z_{t-1} = (1, growth of the composite leading indicator)
- LR of TVTP against constant transitions: 8.94 ($p = 0.011$, 2 restrictions)
- The constant-transition maximum is not a business cycle: a high-mean regime (2.77% per month) that lasts about one month

Transition probabilities driven by the leading indicator



- Top: smoothed probability of the low-growth regime; bottom: the probabilities of staying in each regime, month by month

ATS_ch7_tvtp

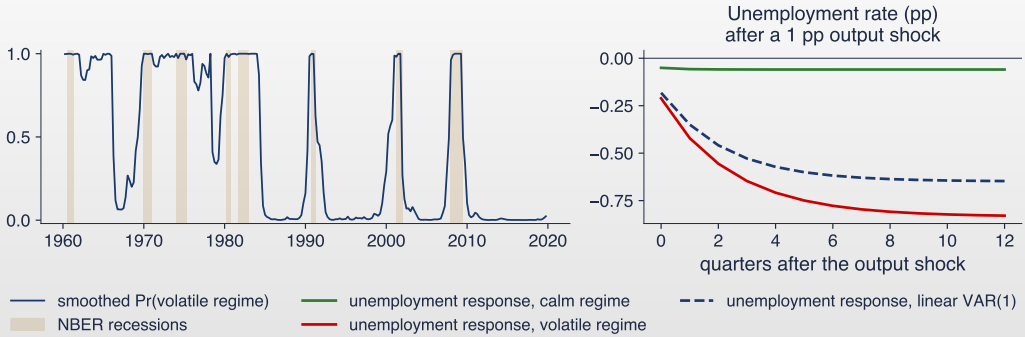
Interpreting TVTP

- Stay in expansion: median 0.988, but down to 0.10 when the leading indicator falls sharply: expansions end when the indicator turns
- Stay in recession falls when the indicator rises (γ slope -1.09): recoveries are announced by the indicator
- QPS against the NBER months 0.197, concordance 88%
- Caveat: the leading indicator was built knowing the NBER dates; a real-time test needs its vintages

MS-VAR and regime-dependent responses

- MSIAH(K)-VAR(p): $y_t = \nu_{S_t} + \sum_{k=1}^{p} A_{k,S_t} y_{t-k} + \Sigma_{S_t}^{1/2} \varepsilon_t$, $y_t \in \mathbb{R}^n$ Krolzig (1997)
  - ν_{S_t}, A_{k,S_t}, Σ_{S_t}: intercept vector, lag matrices and error covariance of the current regime; $\varepsilon_t \sim N(0, I_n)$
  - EM: the M-step is multivariate weighted least squares by regime; $\Sigma_j = \sum_t \hat{\xi}_{j,t|T} \hat{u}_{jt} \hat{u}_{jt}' / \sum_t \hat{\xi}_{j,t|T}$ ($\hat{u}_{jt}$: residuals of regime j)
  - parameters grow as $K(n + n^2 p + n(n+1)/2)$: shrinkage or restricted switching (MSIH) in larger systems (Chapter 5)
- **Regime-dependent impulse responses** Ehrmann, Ellison and Valla (2003): $\partial y_{t+h} / \partial \varepsilon_t$ computed with (A_j, Σ_j), assuming regime j persists over the horizon
  - the full response averages over future regime paths and depends on the current regime probabilities; simulate it
- Switching in Σ alone identifies structural shocks by heteroskedasticity (Chapter 3); Sims and Zha (2006) ask whether US monetary policy switched, and find mainly variance switches

A two-regime VAR for output and unemployment



- MSIAH(2)-VAR(1) for US GDP growth and the change of the unemployment rate, 1960Q1–2019Q4 ($T = 239$, 20 parameters), EM from 12 starts; Cholesky order: output first; responses scaled to a 1 pp output shock

ATS_ch7_msvar

Interpreting the MS-VAR

- The regimes are calm and volatile
  - output shock s.d. 0.42 against 1.02 pp; durations 21.8 and 14.3 quarters
  - the volatile regime holds 83% of quarters before 1984, 17% after, and 100% of NBER quarters
- Unemployment 12 quarters after a 1 pp output shock: -0.06 pp (calm), -0.83 pp (volatile), -0.65 pp (linear VAR)
- Okun's law is regime-dependent
  - in the calm regime output surprises barely move unemployment (correlation of shocks -0.15, against -0.69)
  - a linear VAR averages two transmissions
- In L: -166.8 (MS-VAR) against -222.0 (linear VAR); regime-dependent responses assume the regime persists over the horizon

Regimes in volatility



Lehman Brothers, Times Square, 2007: a calm regime about to end

Photo: David Shankbone (2007); CC BY-SA 3.0; Wikimedia Commons

- Lamoureux and Lastrapes (1990): structural shifts in the variance bias GARCH persistence $\alpha + \beta$ towards 1
- Hamilton and Susmel (1994): SWARCH, an ARCH whose scale switches with a Markov chain; most of the persistence moves into the chain
- MSIH on returns: the simplest volatility-regime model, already a useful benchmark for GARCH Bollerslev (1986)
- Chapter 8 develops realised measures and multivariate GARCH; here: what regimes do to GARCH

The path-dependence problem

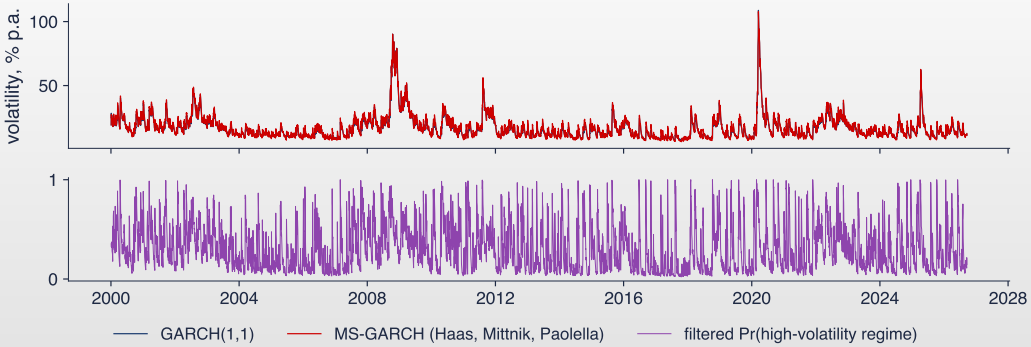
- Naive MS-GARCH: $h_t = \omega_{S_t} + \alpha_{S_t}\varepsilon_{t-1}^2 + \beta_{S_t} h_{t-1}$
  - h_t: the conditional variance; ε_{t-1}: the last return shock; ω, α, β: the GARCH parameters of the regime
  - h_{t-1} is itself regime-dependent
  - h_t depends on the whole path $(S_1, \ldots, S_t)$: K^t histories, so the Hamilton filter cannot be applied ($2^{100} \approx 10^{30}$)
- Gray (1996): collapse the past into $h_{t-1} = \mathbb{E}[\varepsilon_{t-1}^2 \mid Y_{t-2}]$, the variance of the mixture: $h_{j,t} = \omega_j + \alpha_j \varepsilon_{t-1}^2 + \beta_j h_{t-1}$
  - Klaassen (2002): condition on S_t as well, using $\Pr(S_{t-1} \mid S_t, Y_{t-1})$
- Haas, Mittnik and Paolella (2004): K GARCH processes run in parallel, $h_{j,t} = \omega_j + \alpha_j \varepsilon_{t-1}^2 + \beta_j h_{j,t-1}$; the chain picks one
  - no path dependence, exact likelihood, explicit stationarity conditions; asymptotics in Bauwens, Preminger and Rombouts (2010)

S&P 500: GARCH against MS-GARCH

Model	$\alpha+\beta$, regime 1	$\alpha+\beta$, regime 2	p_{ii}	$\ln L$	BIC
GARCH(1,1)	0.981			−9219.9	18475.0
MS-GARCH (HMP)	0.964	0.998	0.959; 0.925	−9119.1	18317.5
MS-GARCH (Gray)	1.000	0.921	0.981; 0.956	−9207.4	18494.1

- Daily log returns (%), January 2000 – September 2026, $T = 6\,714$; Normal innovations, common mean; numerical ML in numpy from several starting values
- HMP regimes: average conditional volatility while in the regime 13.1% and 27.6% per year; expected durations 24 and 13 days

Conditional volatility and the high-volatility regime



- Top: annualised conditional volatility, GARCH(1,1) and MS-GARCH (HMP, mixture variance); bottom: filtered probability of the high-volatility regime

ATS_ch7_msgarch

Interpreting MS-GARCH

- BIC prefers HMP (18317.5) to GARCH (18475.0); the high-volatility regime is active on 21% of days
- Single-regime persistence 0.981; within the regimes 0.964 (calm) and 0.998 (high volatility): regimes do not remove persistence automatically, they move it to where it belongs
- Gray's collapsing fits worse here (-9207.4): the approximation is not innocuous
- Local maxima are frequent: different starts give different regime splits (always report the starting design)

Equity regimes: S&P 500 and BET

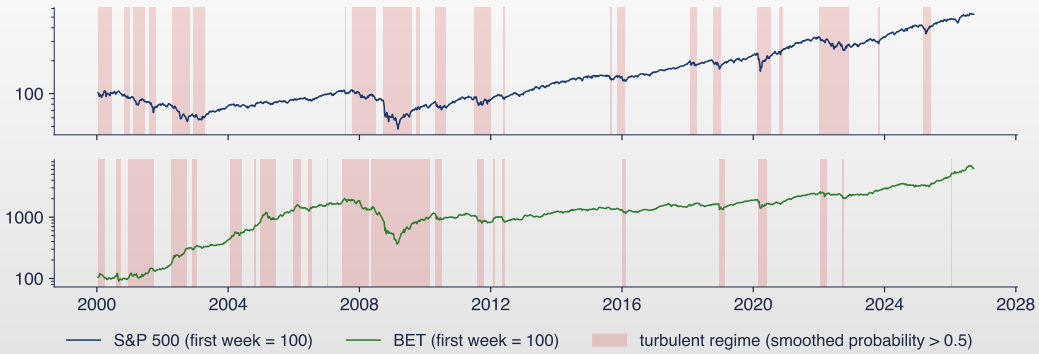


Bucharest Stock Exchange, 2024

Photo: Corina Chitu (2024); CC BY-SA 4.0; Wikimedia Commons

- Weekly log returns, 2000–2026, MSIH(2): $r_t = \mu_{S_t} + \sigma_{S_t}\varepsilon_t$ Maheu and McCurdy (2000), Ang and Timmermann (2012)
- S&P 500 (annualised): calm 16.6% mean, 10.9% volatility; turbulent −20.6%, 28.3%
- BET: calm 20.1%, 13.6%; turbulent 3.2%, 39.9%
- Durations (weeks), calm and turbulent: S&P 500 37 and 14; BET 33 and 12

Turbulent regimes on two markets



- Log index levels (first week = 100); shaded: smoothed probability of the turbulent regime above 0.5; EM from 15 starts and a numerical polish (`statsmodels`: same $\ln L$)

ATS_ch7_bullbear

Interpreting the equity regimes

- Turbulent share: S&P 500 26%, BET 25%; correlation of the two turbulent probabilities 0.48; both turbulent in 15% of weeks
- The regimes are mostly volatility regimes: the turbulent mean is lower but estimated with large error
- $\ln L$: S&P 500 -3019.3 (numpy) and -3019.3 (`statsmodels`); BET -3339.9 and -3339.9
- The BET has its own turbulent episodes (local politics, the 2007–2009 crash), not only imported ones

Regimes and asset allocation, briefly

- Ang and Bekaert (2002)
  - with regime switching, optimal portfolios depend on the current regime probabilities
  - home bias and the value of international diversification change across regimes
- Mean-variance weight of equity against cash, risk aversion $\gamma = 5$: $w = \mu/(\gamma\sigma^2)$; μ, σ^2: weekly mean and variance of the return in the regime
  - S&P 500: 2.79 in the calm regime, -0.51 in the turbulent one, 0.39 with constant moments
  - with one-step mixture moments the weight moves between -0.47 and 2.02
- Caveats: estimation error in regime means is large; out-of-sample gains must be tested (Chapter 1) and transaction costs included
- Market-risk applications (VaR and ES with regimes) are in MFM, Chapter 7

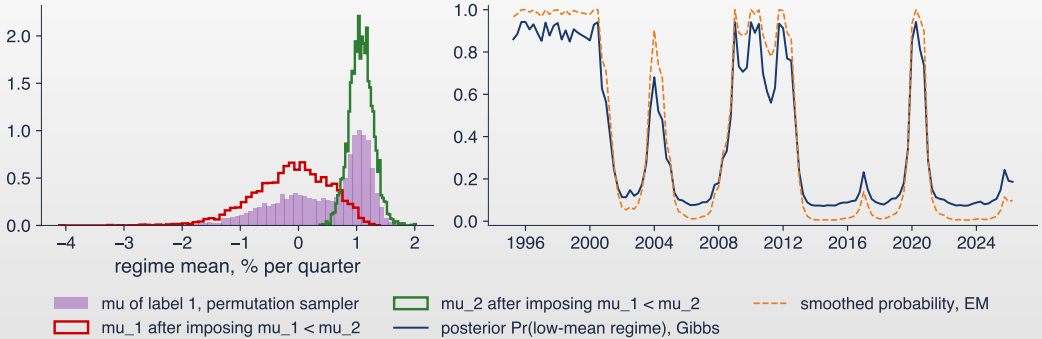
Gibbs sampling with data augmentation

- Treat $S = (S_1, \ldots, S_T)$ as parameters Albert and Chib (1993): given S, the model is a set of regressions; given θ, S is a hidden Markov chain
  - 1. $S \mid \theta, Y$; 2. P $\mid S$: rows $\sim$ Dirichlet$(a_{i\cdot} + n_{i\cdot})$, n_{ij} the transition counts, a_{ij} the prior Dirichlet parameters
  - 3. $\mu_j \mid \sigma_j^2, S, Y \sim$ Normal and $\sigma_j^2 \mid \mu_j, S, Y \sim$ inverse gamma: conjugate updates on the observations of regime j (Chapter 5)
- **FFBS** Chib (1996): draw the whole path at once: $S_T \sim \hat{\xi}_{T|T}$, then $\Pr(S_t = i \mid S_{t+1} = j, Y_t) \propto \hat{\xi}_{i,t|t} p_{ij}$ backwards
  - single-move sampling of S_t given S_{t-1}, S_{t+1} mixes very slowly when regimes are persistent
  - the discrete analogue of the simulation smoother of Chapter 6 Carter and Kohn (1994)
- Priors keep σ_j away from 0 and p_{ii} away from the boundary: the degenerate maxima of ML disappear

Label switching in MCMC, and choosing K

- **Random permutation sampler** Frühwirth-Schnatter (2001): permute the labels at random after each sweep; the sampler then visits all $K!$ modes
  - the unconstrained output shows which parameters separate the regimes; identify afterwards by the constraint that does
  - a bad constraint (e.g. on parameters that barely differ) cuts a mode in half and biases both regimes Celeux, Hurn and Robert (2000)
- Choosing K: marginal likelihood $p(Y \mid K)$ from the Gibbs output Chib (1995), $\ln p(Y) = \ln f(Y \mid \theta^*) + \ln p(\theta^*) - \ln p(\theta^* \mid Y)$
  - the likelihood ordinate comes from the Hamilton filter; the posterior ordinate from reduced Gibbs runs
- Change-point models (Chib 1998) are the same sampler with an upper-triangular P

Romanian GDP growth: two regimes by Gibbs sampling



- Quarterly growth, 1995Q2–2026Q2 ($T = 125$), MSIH(2) without AR term; 5 000 draws after burn-in; left: permutation sampler, right: sampler identified by $\mu_1 < \mu_2$

ATS_ch7_bayes

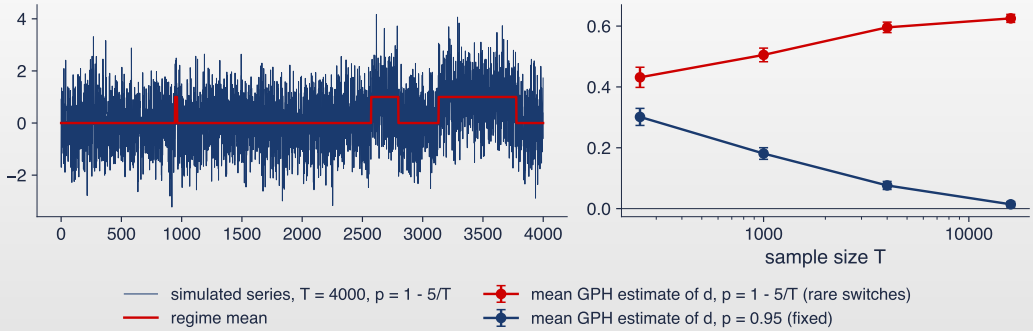
Interpreting the posterior

- Permutation sampler: label 1 carries the larger mean in 49% of draws: the bimodal histogram is the two regimes, not two answers
- Identified: μ_1 −0.11 [−1.26; 0.87], σ_1 3.80; μ_2 1.08 [0.76; 1.41], σ_2 1.57 (90% credible intervals)
- p_{11} 0.85 [0.73; 0.94], p_{22} 0.91; EM: $\mu = 0.03$ and 1.04, $p_{ii} = 0.90$ and 0.94
- The means overlap, the standard deviations do not: the regimes are a volatile era (1995–2000, 2009–2012, 2020) and a stable one; the variance is the better identifying constraint
- Posterior and EM regime probabilities correlate at 0.99; the posterior also carries the parameter uncertainty that the EM path ignores

Regime switching and spurious long memory

- Diebold and Inoue (2001): $y_t = \mu_{S_t} + \varepsilon_t$ with $p_{ii} = 1 - c/T^{\delta}$
  - $c, \delta > 0$: switches become rarer as T grows
  - the variance of partial sums grows like that of an I(d) process, $d > 0$
  - the autocorrelations $\propto \lambda^k$ with $\lambda \to 1$ decay so slowly that, in any finite sample, they look hyperbolic
- Consequences
  - a significant GPH or local Whittle estimate (Chapter 10) does not discriminate fractional integration from occasional breaks or regimes
  - the same holds for GARCH persistence (section Volatility regimes and MS-GARCH) and for unit-root tests with breaks (Chapter 2)
- Which description is "true" matters less than which one forecasts better out of sample

Simulation: GPH estimates of d under regime switching



- $y_t = \mu_{S_t} + \varepsilon_t$, $\mu \in \{0, 1\}$, $\sigma = 1$; mean GPH estimate ($m = T^{0.5}$) over 200 replications, ± 2 standard errors

ATS_ch7_longmem

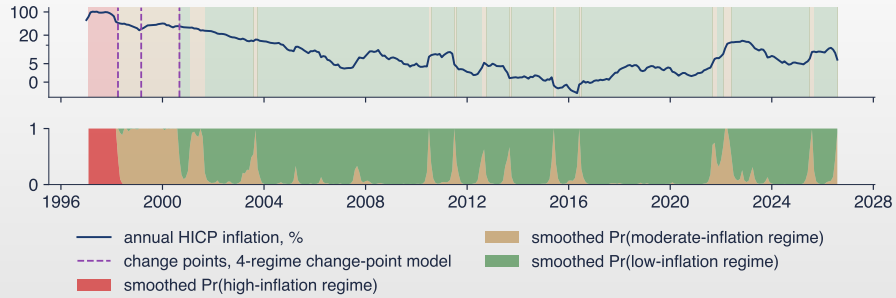
Interpreting the simulation

- Rare switches ($p = 1 - 5/T$): mean $\hat{d} = 0.43, 0.51, 0.60, 0.63$ for $T = 250, 1\,000, 4\,000, 16\,000$: it does not vanish as T grows
- Fixed $p = 0.95$: 0.30, 0.18, 0.08, 0.01: short memory, $\hat{d} \to 0$ as the frequencies used approach zero
- The path on the left has 6 switches in 4000 observations; its autocorrelation is 0.12 at lag 50 and 0.06 at lag 200

Regime switching and structural breaks

- A break (Chapter 2) is a regime that never returns: P upper bidiagonal, $S_1 = 1$ Chib (1998)
  - the Hamilton filter and EM apply unchanged; the transition mask keeps the forbidden transitions at zero
- Recurrent regimes pool information across episodes: the second high-inflation episode reuses the parameters of the first
  - breaks need a new parameter set for every episode, and say nothing about the future
- Compare the two by BIC or marginal likelihood; Garcia and Perron (1996): the US real interest rate is better described by three recurrent regimes than by a stable process

Romanian inflation regimes since 1997



Annual HICP inflation, $100 \ln(P_t / P_{t-12})$, from February 1997 ($T = 355$); MSIH(3)-AR(1), EM from 25 starts; dashed: change points of the 4-regime change-point model; symmetric log scale

ATS_ch7_ro_inflation

Interpreting the inflation regimes

- Regime levels $\nu_j/(1-\phi)$: 68.7%, 17.8% and 2.9%; shock s.d. 10.58, 1.70 and 0.45 pp; $\phi = 0.971$: after 2001 the two lower regimes differ mainly in volatility
- The volatile regime returns in the 2010 VAT increase, in 2022 and in 2025; filtered probabilities in August 2026 (inflation 6.1%): 0.00, 0.93, 0.07
- BIC prefers change points: 3 segments 887.0, 4 segments 897.5, Markov switching 934.1; the change points of the 4-segment model: April 1998, March 1999, September 2000
- Reading: disinflation was a one-way process
  - after 2000 a single, very persistent segment absorbs the 2022 and 2025 surges as shocks
  - whether they are a recurrent regime is decided out of sample

Romania's exchange-rate regimes

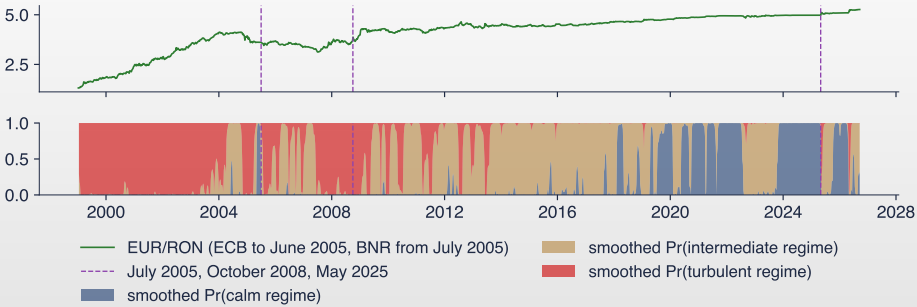


National Bank of Romania, Bucharest, 2014

Photo: Crislia (2014); CC BY-SA 4.0; Wikimedia Commons

- 1999–2005: crawling depreciation of the old leu under a managed float
- July 2005: redenomination (10 000 ROL = 1 RON); August 2005: inflation targeting with a managed float; capital account fully liberalised in 2006
- Episodes: October 2008, 2011–2012, March 2020, May 2025 (presidential election and fiscal stress)
- Model: weekly log changes, MSIH(3) ordered by volatility; regimes are policy outcomes, not the announced regime

EUR/RON volatility regimes, 1999–2026



- Top: EUR/RON (ECB reference rate to June 2005, BNR from July 2005); bottom: smoothed probabilities of the calm, intermediate and turbulent regimes; $T = 1\,445$ weeks

ATS_ch7_eurron

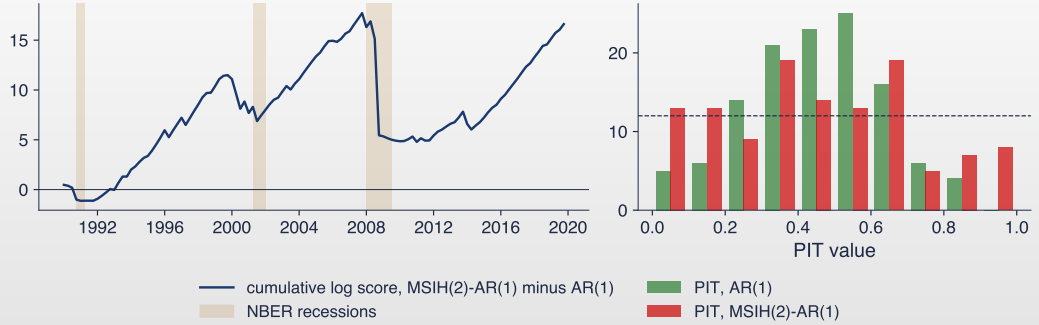
Interpreting the EUR/RON regimes

- Weekly s.d. by regime: 0.07, 0.39 and 1.51%; expected durations 16, 17 and 20 weeks
- Before July 2005 the turbulent regime holds 87% of weeks (the steady depreciation of the old leu, mean 0.25% per week), after it 18%; the calm regime goes from 2% to 25%
- Turbulent probability peaks at 1.00 in October 2008–March 2009 and at 1.00 in spring 2025; the largest 2025 weekly change, 2.76%, in the week ending 9 May 2025
- EUR/RON averaged 4.9773 in April 2025 and stood at 5.2644 at the end of the sample: a level shift followed by a return to calm, as a managed float predicts

Forecasts from a switching model

- Regime forecast: $\hat{\xi}_{T+h|T} = (\mathrm{P}')^h \hat{\xi}_{T|T}$, converging to $\boldsymbol{\pi}$ at the rate λ^h
  - one step: $f(y_{T+1} \mid Y_T) = \sum_j \hat{\xi}_{j,T+1|T}\, N(x'_{T+1}\beta_j, \sigma_j^2)$, a **mixture**: skewed, fat-tailed, possibly bimodal
- Several steps with AR terms: the mean is not linear in past regimes (MSM) or needs the regime path; simulate paths of (S, y)
- Evaluation (Chapter 1): point forecasts by RMSE and Diebold–Mariano; densities by log score, PIT and CRPS; regime probabilities by QPS
  - QPS $= \frac{2}{T}\sum_t (\hat{p}_t - d_t)^2$ Diebold and Rudebusch (1989): the Brier score of a recession probability $\hat{p}_t$ against the NBER indicator d_t (1 in recession); 0 is perfect, 2 the worst
- Clements and Krolzig (1998): MS models rarely beat linear AR in point forecasts of GDP; their value is in densities and in turning-point probabilities

US GDP: one-step density forecasts, 1990–2019



- Expanding window from 1947Q2, 120 quarters, AR(1) re-estimated each quarter, MSIH(2)-AR(1) every 4 quarters (EM, warm start); left: cumulative log-score difference; right: PIT histograms (dashed: uniform)

ATS_ch7_forecast

Interpreting the forecast comparison

- RMSE: AR(1) 0.554, MSIH(2)-AR(1) 0.558: the point forecasts are almost the same, as Clements and Krolzig found
- Mean log score: -1.028 against -0.890; Diebold–Mariano on the difference 1.16 ($p = 0.24$)
- The gain comes from the variance regime: the AR(1) density is too wide after 1984 (PIT piled in the middle, KS $p < 0.001$); the MS density is closer to uniform but not calibrated either (KS $p = 0.01$)
- Cumulative log-score gain before 2008: 17.72; from 2008: -1.10: the regime model pays in calm times and loses in the crisis

An open question

- Can a regime model date Romanian recessions and inflation regimes in real time better than simple rules, and is the 2025 episode a new regime?
  - formal: H_0: equal QPS of the filtered regime probability and of a two-negative-quarters rule against a reference chronology, on pre-registered quarters
  - falsified by a significant difference on quarters the model never saw, with GDP vintages
- Why it matters: Romania has no official business-cycle dating committee; policy reacts to regimes it can see only with delay
  - literature to start from: Hamilton (1989), Chauvet and Piger (2008), Clements and Krolzig (1998), Hamilton (2016)

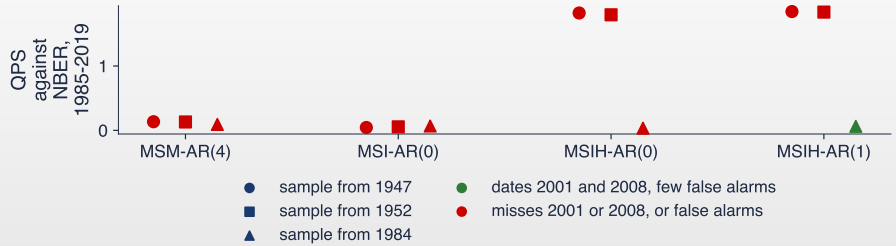
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers that date business cycles of Central and Eastern European countries with Markov-switching models; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Which specification (MSM, MSIH, TVTP with which leading indicator) should separate recessions from volatility eras in Romanian GDP?`
  - **code and replication**: ask for a Hamilton filter, then reproduce a known number first (Hamilton's log-likelihood -181.26)
  - **robustness and critique**: `Act as a hostile referee: list the ways a regime dating could be an artefact of the specification or of the starting values.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

Required checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The likelihood is the global maximum among many starts, and the chosen regime is not degenerate
- Real-time claims use filtered probabilities and data vintages, never smoothed probabilities
- The number of regimes is tested by bootstrap or chosen by a criterion fixed in advance, not by a χ^2 count
- The meaning of each regime is checked against external events, not assumed from its label

Mini-case: sensitivity of the recession dating to the specification



- US GDP growth: four specifications × three sample starts, all ending in 2019Q4; QPS of the smoothed low-mean-regime probability against the NBER quarters, 1985–2019
- QPS from 0.034 to 1.846 (a constant probability scores 0.145)
  - 4 variants flag most quarters: their low-mean regime is the Great Moderation
  - 1 of 12 dates both 2001 and 2008 without false alarms: an AI summary that reports one variant as "the" dating is wrong

ATS_ch7_hamilton

Project idea

- **A regime-based business-cycle chronology for Romania**: replicate first, then extend
  - replicate: Hamilton's Table I on his data (log-likelihood -181.26) and the Filardo TVTP model (-586.20)
  - extend: Romanian GDP and industrial production, MSM against MSIH, TVTP with the ESI as leading indicator, Bayesian K selection, real-time filtered probabilities
  - pre-register: sample, specifications, starting-value design, the reference chronology and the QPS comparison
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- The Hamilton filter and the Kim smoother are exact Bayesian recursions for a discrete latent state
- EM and numerical ML find local and degenerate maxima: many starts, then judgement
- The number of regimes is a non-standard testing problem: bootstrap, Garcia, CHP or information criteria
- TVTP, MS-VAR and MS-GARCH make the regimes economic; Gibbs sampling makes the uncertainty visible
- Regimes, breaks and long memory are observationally close; forecasts decide

Self-assessment

Questions

- Why does the Kim smoother need $\hat{\xi}_{t+1|t}$ in the denominator?
- Why does Hamilton's MSM-AR(4) need 32 states?
- Which three conditions of the LR asymptotics fail when testing one regime against two?
- Why is the naive MS-GARCH likelihood path-dependent?
- Why can a regime-switching mean produce a positive GPH estimate?

Next:

- Advanced volatility modelling
- Realised measures, HAR and multivariate GARCH

Appendix: EM for the Markov chain, step by step

- Expected complete-data log-likelihood:
  $Q(\theta \mid \theta^{(m)}) = \sum_j \hat{\xi}_{j,1|T} \ln \rho_j + \sum_{t \geq 2} \sum_{i,j} \hat{\xi}_{ij,t|T} \ln p_{ij} + \sum_t \sum_j \hat{\xi}_{j,t|T} \ln \eta_{jt}$
- Maximise $\sum_{t,j} \hat{\xi}_{ij,t|T} \ln p_{ij}$ subject to $\sum_j p_{ij} = 1$: Lagrangian $\Rightarrow p_{ij} = \sum_t \hat{\xi}_{ij,t|T} / \sum_t \sum_k \hat{\xi}_{ik,t|T}$
- and $\sum_k \hat{\xi}_{ik,t|T} = \hat{\xi}_{i,t-1|T}$; the Gaussian part gives weighted least squares
- Monotonicity: $\ln L(\theta) = Q(\theta \mid \theta^{(m)}) - H(\theta \mid \theta^{(m)})$, and H is maximised at $\theta^{(m)}$ (Gibbs inequality) Dempster, Laird and Rubin (1977)

Back

Appendix: the Kim smoother, the Markov step

- $\Pr(S_t = i \mid S_{t+1} = j, Y_T) = \dfrac{f(y_{t+1:T} \mid S_t = i, S_{t+1} = j, Y_t)\Pr(S_t = i \mid S_{t+1} = j, Y_t)}{f(y_{t+1:T} \mid S_{t+1} = j, Y_t)}$
- If $y_{t+1:T}$ depends on S_t only through S_{t+1} (true on the expanded state), the density ratio is 1
- Then $\Pr(S_t = i \mid S_{t+1} = j, Y_t) = \hat{\xi}_{i,t|t} p_{ij} / \hat{\xi}_{j,t+1|t}$; multiply by $\hat{\xi}_{j,t+1|T}$ and sum over j
- With a continuous state (MS state space, Chapter 6) the ratio is not 1: Kim's smoother is then an approximation

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References (1/4)

- Albert, J. H., & Chib, S. (1993). Bayes inference via Gibbs sampling of autoregressive time series subject to Markov mean and variance shifts. *Journal of Business & Economic Statistics*, 11(1), 1–15. doi:10.1080/07350015.1993.10509929
- Ang, A., & Bekaert, G. (2002). International asset allocation with regime shifts. *The Review of Financial Studies*, 15(4), 1137–1187. doi:10.1093/rfs/15.4.1137
- Ang, A., & Timmermann, A. (2012). Regime changes and financial markets. *Annual Review of Financial Economics*, 4, 313–337. doi:10.1146/annurev-financial-110311-101808
- Baum, L. E., Petrie, T., Soules, G., & Weiss, N. (1970). A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1), 164–171. doi:10.1214/aoms/1177697196
- Bauwens, L., Preminger, A., & Rombouts, J. V. K. (2010). Theory and inference for a Markov switching GARCH model. *The Econometrics Journal*, 13(2), 218–244. doi:10.1111/j.1368-423X.2009.00307.x
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- Carrasco, M., Hu, L., & Ploberger, W. (2014). Optimal test for Markov switching parameters. *Econometrica*, 82(2), 765–784. doi:10.3982/ECTA8609
- Carter, C. K., & Kohn, R. (1994). On Gibbs sampling for state space models. *Biometrika*, 81(3), 541–553. doi:10.1093/biomet/81.3.541
- Celeux, G., Hurn, M., & Robert, C. P. (2000). Computational and inferential difficulties with mixture posterior distributions. *Journal of the American Statistical Association*, 95(451), 957–970. doi:10.1080/01621459.2000.10474285
- Chauvet, M., & Piger, J. (2008). A comparison of the real-time performance of business cycle dating methods. *Journal of Business & Economic Statistics*, 26(1), 42–49. doi:10.1198/073500107000000296
- Chib, S. (1996). Calculating posterior distributions and modal estimates in Markov mixture models. *Journal of Econometrics*, 75(1), 79–97. doi:10.1016/0304-4076(95)01770-4
- Chib, S. (1998). Estimation and comparison of multiple change-point models. *Journal of Econometrics*, 86(2), 221–241. doi:10.1016/S0304-4076(97)00115-2

References (2/4)

- Chib, S. (1995). Marginal likelihood from the Gibbs output. *Journal of the American Statistical Association*, 90(432), 1313–1321. doi:10.1080/01621459.1995.10476635
- Cho, J. S., & White, H. (2007). Testing for regime switching. *Econometrica*, 75(6), 1671–1720. doi:10.1111/j.1468-0262.2007.00809.x
- Clements, M. P., & Krolzig, H.-M. (1998). A comparison of the forecast performance of Markov-switching and threshold autoregressive models of US GNP. *The Econometrics Journal*, 1(1), C47–C75. doi:10.1111/1368-423X.11004
- Davies, R. B. (1987). Hypothesis testing when a nuisance parameter is present only under the alternative. *Biometrika*, 74(1), 33–43. doi:10.1093/biomet/74.1.33
- Dempster, A. P., Laird, N. M., & Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1), 1–22. doi:10.1111/j.2517-6161.1977.tb01600.x
- Diebold, F. X., & Inoue, A. (2001). Long memory and regime switching. *Journal of Econometrics*, 105(1), 131–159. doi:10.1016/S0304-4076(01)00073-2
- Diebold, F. X., Lee, J.-H., & Weinbach, G. C. (1994). Regime switching with time-varying transition probabilities. In C. P. Hargreaves (Ed.), *Nonstationary Time Series Analysis and Cointegration* (pp. 283–302). Oxford University Press. doi:10.1093/oso/9780198773917.003.0010
- Diebold, F. X., & Rudebusch, G. D. (1989). Scoring the leading indicators. *The Journal of Business*, 62(3), 369–391. doi:10.1086/296467
- Ehrmann, M., Ellison, M., & Valla, N. (2003). Regime-dependent impulse response functions in a Markov-switching vector autoregression model. *Economics Letters*, 78(3), 295–299. doi:10.1016/S0165-1765(02)00256-2
- Filardo, A. J. (1994). Business-cycle phases and their transitional dynamics. *Journal of Business & Economic Statistics*, 12(3), 299–308. doi:10.1080/07350015.1994.10524545
- Frühwirth-Schnatter, S. (2001). Markov chain Monte Carlo estimation of classical and dynamic switching and mixture models. *Journal of the American Statistical Association*, 96(453), 194–209. doi:10.1198/016214501750333063
- Frühwirth-Schnatter, S. (2006). *Finite Mixture and Markov Switching Models*. Springer. doi:10.1007/978-0-387-35768-3

References (3/4)

- Garcia, R. (1998). Asymptotic null distribution of the likelihood ratio test in Markov switching models. *International Economic Review*, 39(3), 763–788. doi:10.2307/2527399
- Garcia, R., & Perron, P. (1996). An analysis of the real interest rate under regime shifts. *The Review of Economics and Statistics*, 78(1), 111–125. doi:10.2307/2109851
- Goldfeld, S. M., & Quandt, R. E. (1973). A Markov model for switching regressions. *Journal of Econometrics*, 1(1), 3–15. doi:10.1016/0304-4076(73)90002-X
- Gray, S. F. (1996). Modeling the conditional distribution of interest rates as a regime-switching process. *Journal of Financial Economics*, 42(1), 27–62. doi:10.1016/0304-405X(96)00875-6
- Haas, M., Mittnik, S., & Paolella, M. S. (2004). A new approach to Markov-switching GARCH models. *Journal of Financial Econometrics*, 2(4), 493–530. doi:10.1093/jjfinec/nbh020
- Hamilton, J. D. (1990). Analysis of time series subject to changes in regime. *Journal of Econometrics*, 45(1–2), 39–70. doi:10.1016/0304-4076(90)90093-9
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384. doi:10.2307/1912559
- Hamilton, J. D. (2016). Macroeconomic regimes and regime shifts. In J. B. Taylor & H. Uhlig (Eds.), *Handbook of Macroeconomics* (Vol. 2A, pp. 163–201). Elsevier. doi:10.1016/bs.hesmac.2016.03.004
- Hamilton, J. D., & Susmel, R. (1994). Autoregressive conditional heteroskedasticity and changes in regime. *Journal of Econometrics*, 64(1–2), 307–333. doi:10.1016/0304-4076(94)90067-1
- Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press. doi:10.2307/j.ctv14jx6sm
- Hansen, B. E. (1992). The likelihood ratio test under nonstandard conditions: Testing the Markov switching model of GNP. *Journal of Applied Econometrics*, 7(S1), S61–S82. doi:10.1002/jae.3950070506
- Harding, D., & Pagan, A. (2002). Dissecting the cycle: A methodological investigation. *Journal of Monetary Economics*, 49(2), 365–381. doi:10.1016/S0304-3932(01)00108-8

References (4/4)

- Hathaway, R. J. (1985). A constrained formulation of maximum-likelihood estimation for normal mixture distributions. *The Annals of Statistics*, 13(2), 795–800. doi:10.1214/aos/1176349557
- Kim, C.-J. (1994). Dynamic linear models with Markov-switching. *Journal of Econometrics*, 60(1–2), 1–22. doi:10.1016/0304-4076(94)90036-1
- Kim, C.-J., & Nelson, C. R. (1999). *State-Space Models with Regime Switching: Classical and Gibbs-Sampling Approaches with Applications*. MIT Press. doi:10.7551/mitpress/6444.001.0001
- Klaassen, F. (2002). Improving GARCH volatility forecasts with regime-switching GARCH. *Empirical Economics*, 27(2), 363–394. doi:10.1007/s001810100100
- Krolzig, H.-M. (1997). *Markov-Switching Vector Autoregressions: Modelling, Statistical Inference, and Application to Business Cycle Analysis*. Springer. doi:10.1007/978-3-642-51684-9
- Lamoureux, C. G., & Lastrapes, W. D. (1990). Persistence in variance, structural change, and the GARCH model. *Journal of Business & Economic Statistics*, 8(2), 225–234. doi:10.1080/07350015.1990.10509794
- Maheu, J. M., & McCurdy, T. H. (2000). Identifying bull and bear markets in stock returns. *Journal of Business & Economic Statistics*, 18(1), 100–112. doi:10.1080/07350015.2000.10524851
- Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., et al. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257–286. doi:10.1109/5.18626
- Sims, C. A., & Zha, T. (2006). Were there regime switches in U.S. monetary policy? *American Economic Review*, 96(1), 54–81. doi:10.1257/000282806776157678
- Stephens, M. (2000). Dealing with label switching in mixture models. *Journal of the Royal Statistical Society: Series B*, 62(4), 795–809. doi:10.1111/1467-9868.00265